

DATABASE MANAGEMENT SYSTEMS (CS301)

Stored Procedures

Learning Objectives

Understand stored procedures, their syntax, advantages, parameters, execution methods, and practical database applications.

Introduction to Stored Procedures

A Stored Procedure is a precompiled collection of SQL statements stored inside the database. It can be executed whenever required to perform specific database operations.

Why Use Stored Procedures?

- Reuse SQL code
- Improve performance
- Reduce network traffic
- Enhance security
- Simplify application development

Basic Syntax

```
CREATE PROCEDURE ProcedureName AS BEGIN SQL Statements END;
```

Example Procedure

```
CREATE PROCEDURE GetStudents AS BEGIN SELECT * FROM Student; END;
```

Executing a Stored Procedure

```
EXEC GetStudents;
```

Stored Procedure with Parameters

Parameters allow values to be passed into a stored procedure at runtime.

Parameterized Example

```
CREATE PROCEDURE GetStudentByID @StudentID INT AS BEGIN SELECT * FROM Student  
WHERE StudentID = @StudentID; END;
```

Input and Output Parameters

Input Parameters: Used to send values into a procedure.

Output Parameters: Used to return values from a procedure.

Advantages of Stored Procedures

- Faster execution
- Improved security
- Code reusability

- Easier maintenance
- Reduced development effort

Applications

- Banking Systems
- Payroll Processing
- Inventory Management
- University Management Systems
- E-Commerce Applications

Difference Between Stored Procedures and Functions

Stored Procedure: Can perform multiple operations and may not return a value.

Function: Must return a value and is generally used in expressions.

Exam Important Questions

1. Define Stored Procedure.
2. Explain the syntax of a stored procedure.
3. Discuss advantages of stored procedures.
4. Explain parameterized procedures.
5. Differentiate procedures and functions.

Summary

Stored Procedures are powerful database objects that improve performance, security, and code reusability. They are widely used in enterprise database applications.